

# Spotify 2023 Data Analysis

Power BI Report: Data Cleaning, DAX Measures and Visual Analysis

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**Institution:** Chandigarh University

**Subject:** Business Intelligence

**Tool:** Microsoft Power BI Desktop (with Power Query Editor)

**Dataset:** spotify-2023.csv

**Date:** 28 September 2026

# 1. Objective

The objective of this report is to import the Spotify 2023 songs dataset into Power BI, clean and prepare it using Power Query, create calculated measures with DAX, and build interactive visuals that explore which tracks and artists attract the most streams, how audio features relate to streaming popularity, and how songs perform across Spotify, Apple and Deezer.

# 2. Dataset Overview

The source file **spotify-2023.csv** contains 24 columns and 953 rows before cleaning. Key fields used in this report:

- **track\_name**, **artist(s)\_name**, **artist\_count**: track and artist identification.
- **released\_year**, **released\_month**, **released\_day**: release date components.
- **streams**: total Spotify streams for the track.
- **in\_spotify\_playlists** / **in\_spotify\_charts**, **in\_apple\_playlists** / **in\_apple\_charts**, **in\_deezer\_playlists** / **in\_deezer\_charts**, **in\_shazam\_charts**: platform presence.
- **bpm**, **key**, **mode**, **danceability\_%**, **valence\_%**, **energy\_%**, **acousticness\_%**, **instrumentalness\_%**, **liveness\_%**, **speechiness\_%**: audio characteristics.

# 3. Data Import

The CSV file was loaded through **Get data** in Power BI Desktop. The preview dialog used file origin *1252: Western European (Windows)*, the *Comma* delimiter and data type detection based on the first 200 rows. **Transform Data** was selected (instead of Load) so the data could be cleaned in Power Query before use.

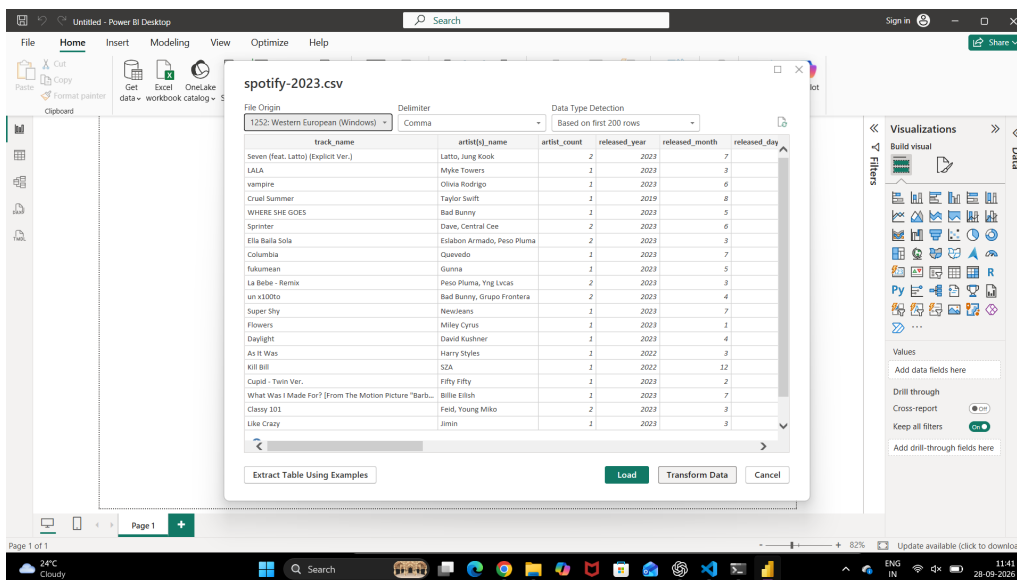


Figure 1: Importing spotify-2023.csv and choosing Transform Data

# 4. Data Cleaning in Power Query

The following applied steps were performed in the Power Query Editor, in this order:

- **Promoted Headers**: the first row was used as the column names.

- **Changed Type:** text columns (track\_name, artist(s)\_name) were set to Text and numeric columns such as artist\_count set to whole numbers (Int64).
- **Removed Duplicates:** duplicate track records were removed, reducing the table from 953 rows to 943 rows.
- **Cleaned Text:** Transform > Format > Clean was applied to remove non-printable characters from text values.
- **Removed Blank Rows:** Home > Remove Rows > Remove Blank Rows removed any empty rows.

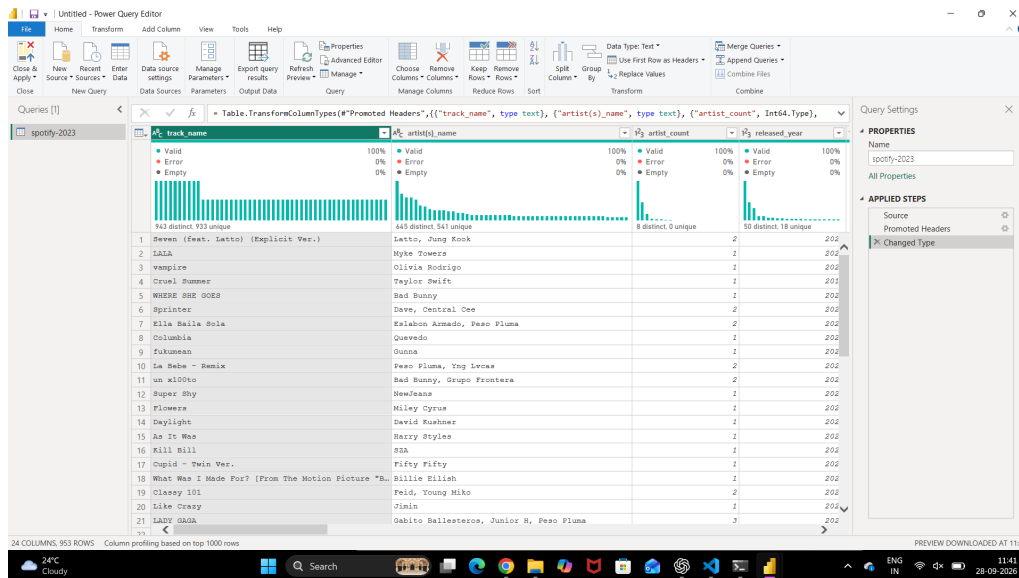


Figure 2: Promoted Headers and Changed Type (953 rows)

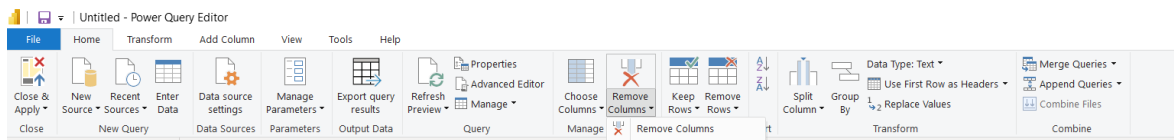


Figure 3: Remove Columns option in the Home ribbon of Power Query Editor

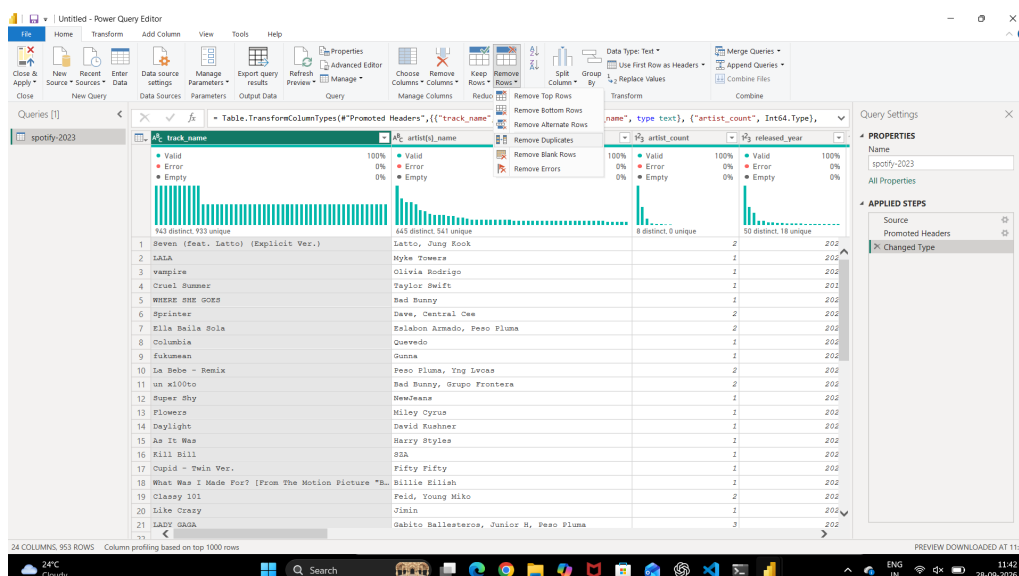


Figure 4: Home > Remove Rows > Remove Duplicates

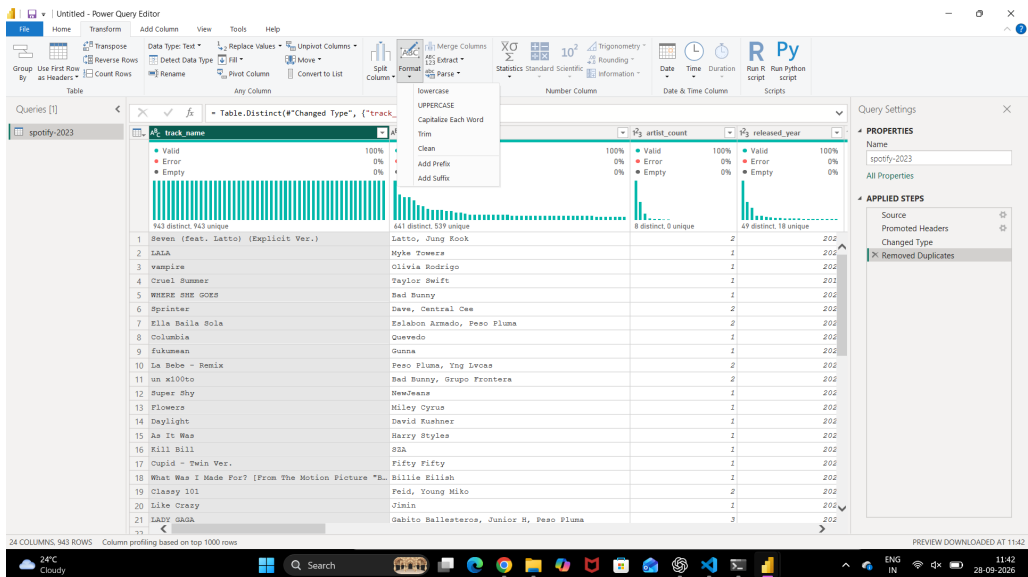


Figure 5: Transform > Format menu (lowercase, UPPERCASE, Capitalize Each Word, Trim, Clean)

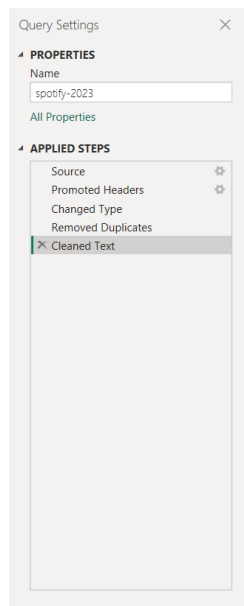


Figure 6: Applied Steps after the Cleaned Text step

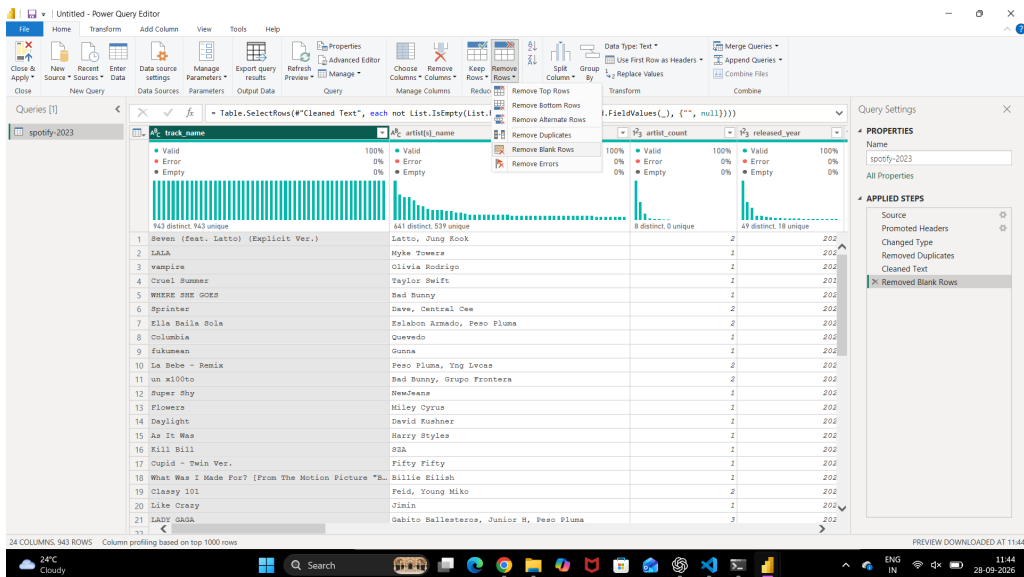


Figure 7: Remove Blank Rows applied; the final query has 943 rows and 24 columns

After cleaning, the query was closed and applied, loading the prepared table **spotify-2023** into the Power BI report view.

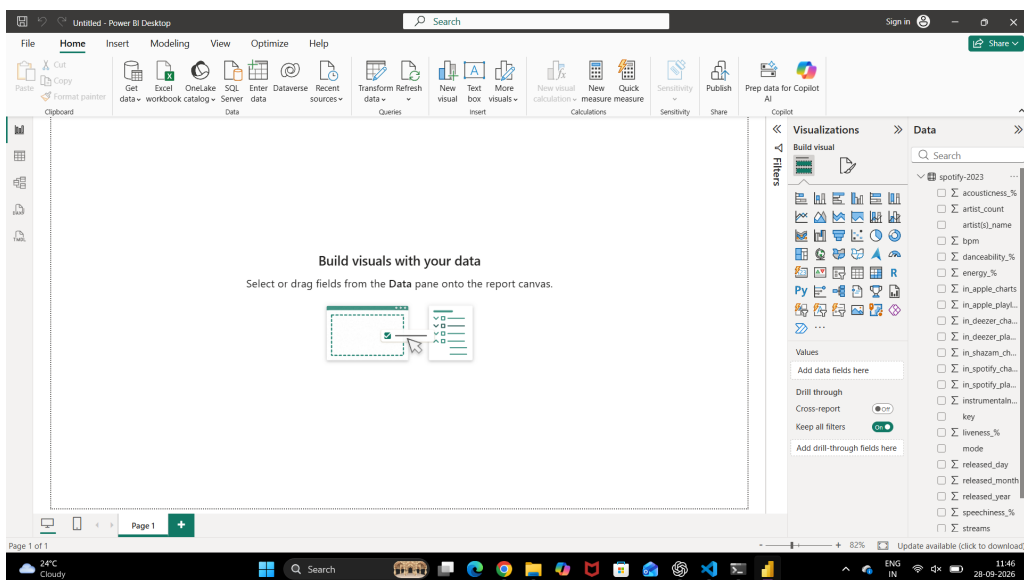


Figure 8: Report canvas with the spotify-2023 fields available in the Data pane

## 5. DAX Measures

New measures were created using **Modeling > New measure**. Each measure uses the DAX AVERAGE function on a column of the spotify-2023 table.

| Measure name         | DAX formula                                  |
|----------------------|----------------------------------------------|
| Average Streams      | AVERAGE('spotify-2023'[streams])             |
| Average Energy       | AVERAGE('spotify-2023'[energy_%])            |
| Avg Apple Playlists  | AVERAGE('spotify-2023'[in_apple_playlists])  |
| Avg deezer Playlists | AVERAGE('spotify-2023'[in_deezer_playlists]) |

Further measures named **Avg Spotify Playlists**, **Average Danceability** and **Total Streams** were also created and appear in the Data pane; Total Streams is the value used on the axes of the visuals in the next section.

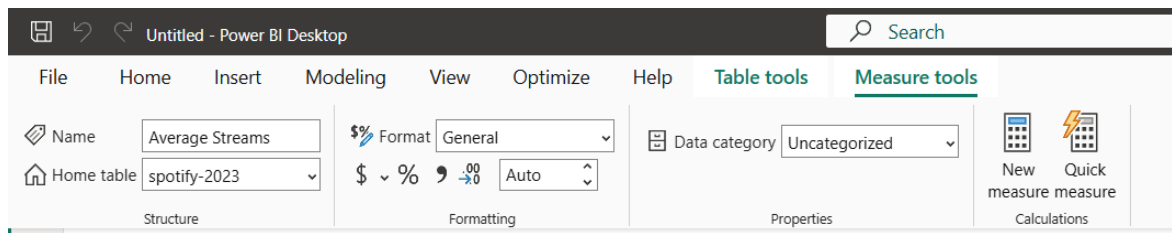


Figure 9: Naming the measure and setting its home table and format

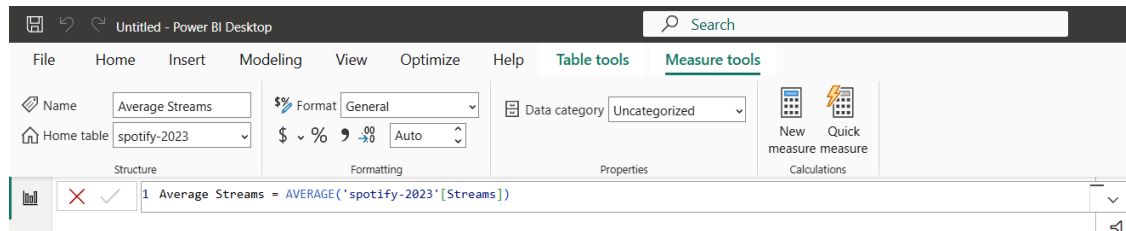


Figure 10: Average Streams measure formula

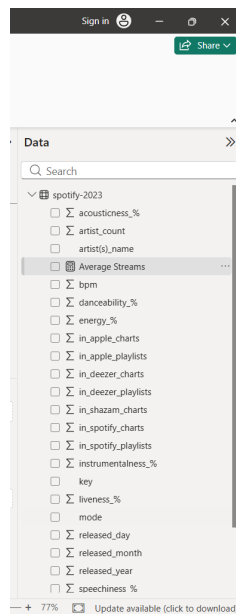


Figure 11: Average Streams appears in the Data pane with the calculator icon

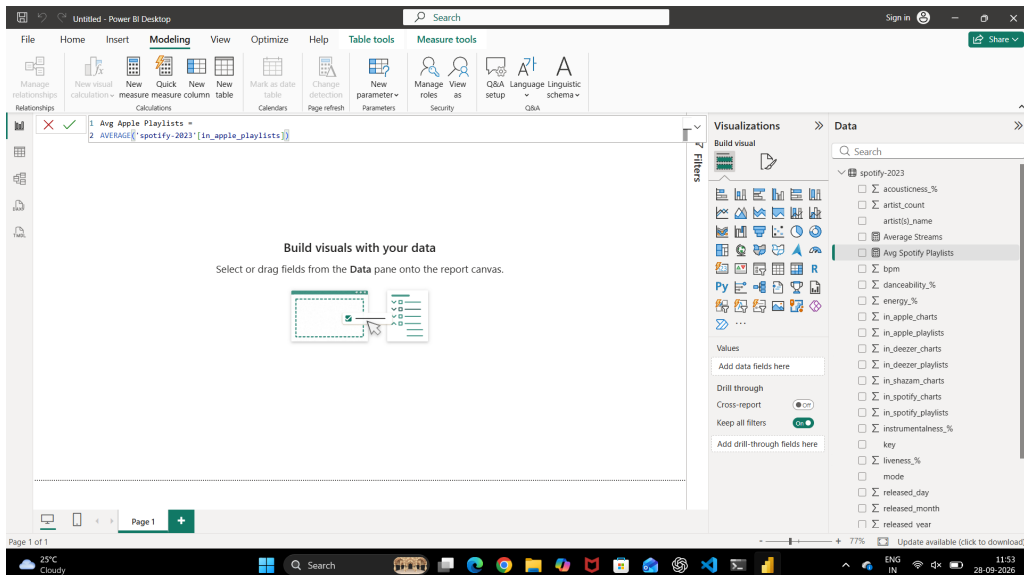


Figure 12: Avg Apple Playlists measure

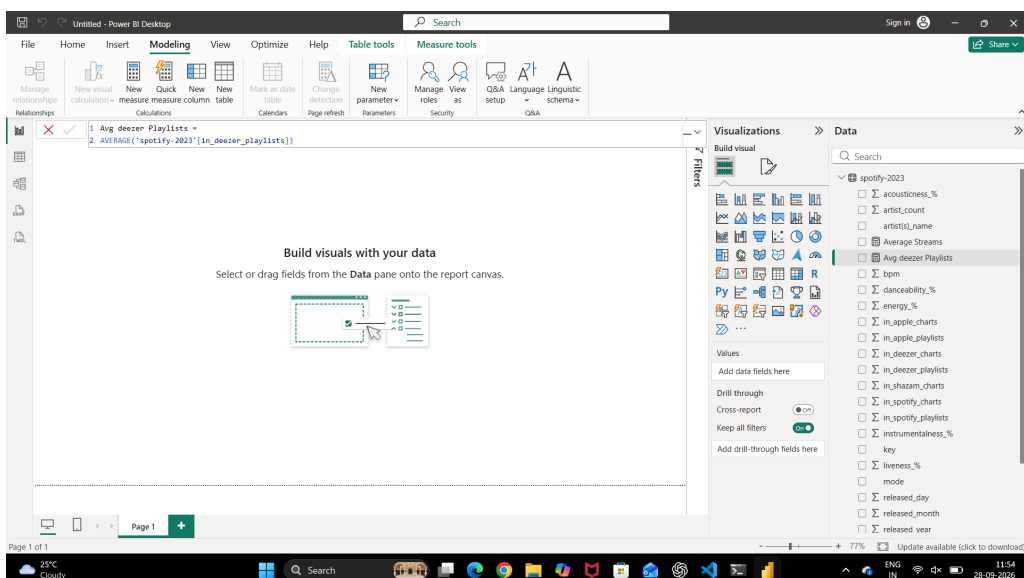


Figure 13: Avg deezer Playlists measure

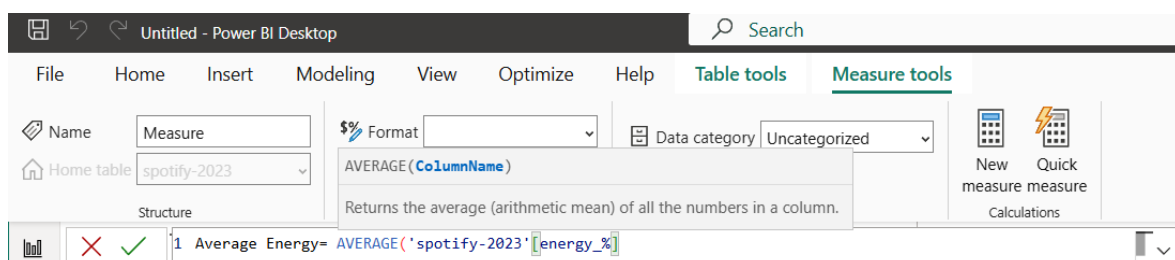


Figure 14: Average Energy measure

## 6. Report Pages and Visual Analysis

### 6.1 Page 1: Top Tracks by Streams

A clustered bar chart plots **Total Streams** (X-axis) against **track\_name** (Y-axis). The chart shows that **Blinding Lights** and **Shape of You** lead the dataset at roughly 3.5 to 4 billion streams each, followed by **Someone You Loved**, **Dance Monkey** and **Sunflower (Spider-Man: Into the Spider-Verse)**.

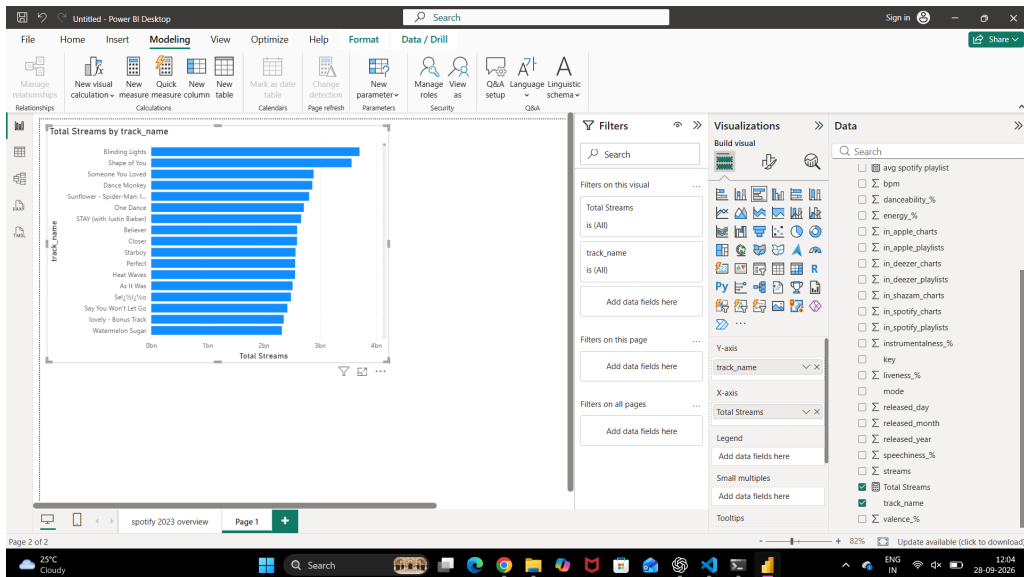


Figure 15: Total Streams by track\_name

An **artist(s)\_name slicer** was then added so the bar chart can be filtered by artist. With *Bad Bunny* selected, the chart lists his tracks, led by Titi Me Pregunto and Yonaguni at about 1 billion streams each, followed by Efecto and Moscow Mule.

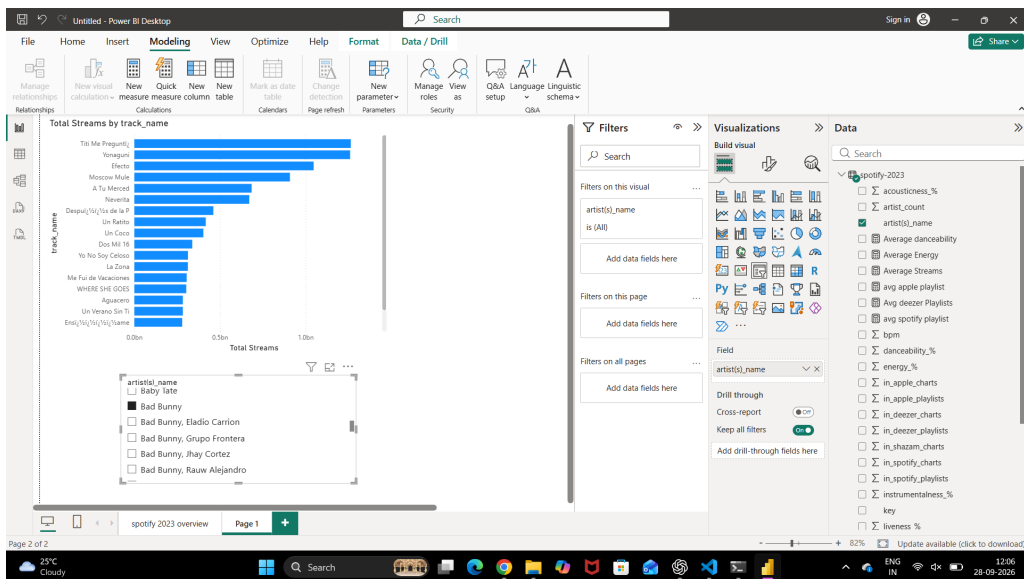


Figure 16: Track chart filtered by the artist slicer (Bad Bunny)

## 6.2 Page 2: Audio Features vs Streams (Bubble/Scatter Charts)

Scatter (bubble) charts were built with **Total Streams** on the Y-axis, an audio feature on the X-axis, **artist(s)\_name** in the Legend and **Sum of streams** as the bubble Size. The first chart uses **danceability\_%**; the same design was duplicated for **energy\_%** and **liveness\_%**. Across all three charts a small number of artists sit far above the rest at roughly 13 to 14 billion total streams, while the majority of artists cluster near the bottom-left with far fewer streams.

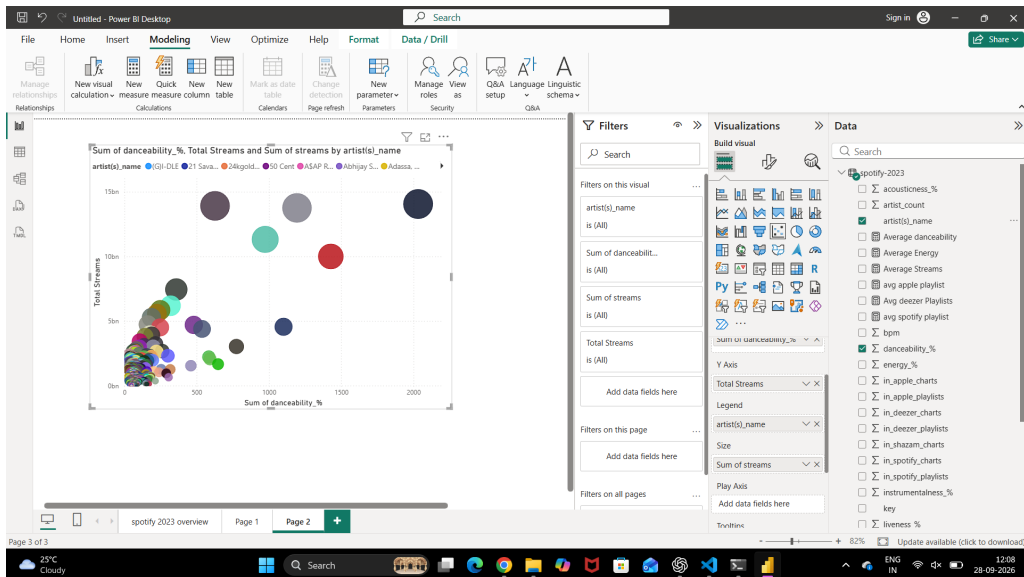


Figure 17: Danceability vs Total Streams by artist

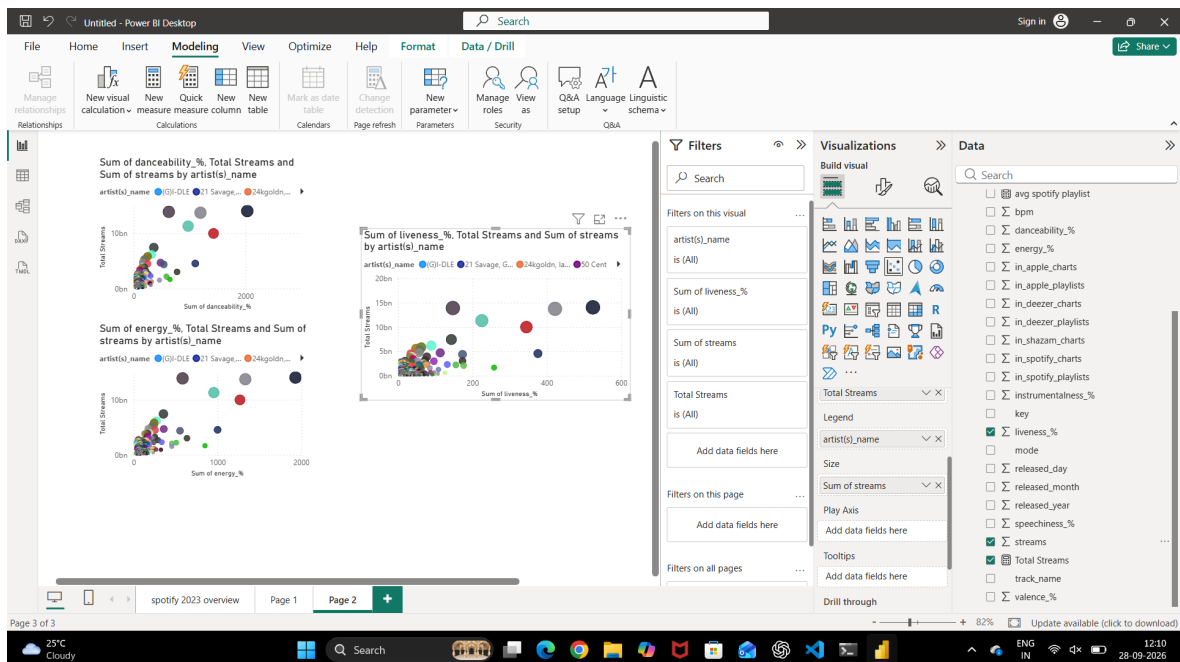


Figure 18: Energy and liveness bubble charts alongside the danceability chart

**Note:** the X-axis uses *Sum of* the feature percentage, so an artist with many songs scores higher simply because more values are added. Using the **Average Danceability** or **Average Energy** measures on the X-axis would compare the actual audio profile of each artist more fairly.

### 6.3 Page 3: Platform Presence by Artist (Multi-row Card)

A multi-row card with **artist(s)\_name** as the category displays the sums of `in_apple_charts`, `in_apple_playlists`, `in_deezer_charts`, `in_deezer_playlists`, `in_spotify_charts` and `in_spotify_playlists`. For example, (G)I-DLE has 881 Spotify playlist placements, 242 Apple charts and 24 Apple playlist entries, while 24kgoldn, Iann Dior shows about 13K Spotify playlist placements, showing how much stronger Spotify playlist reach is than the other platforms for that artist.

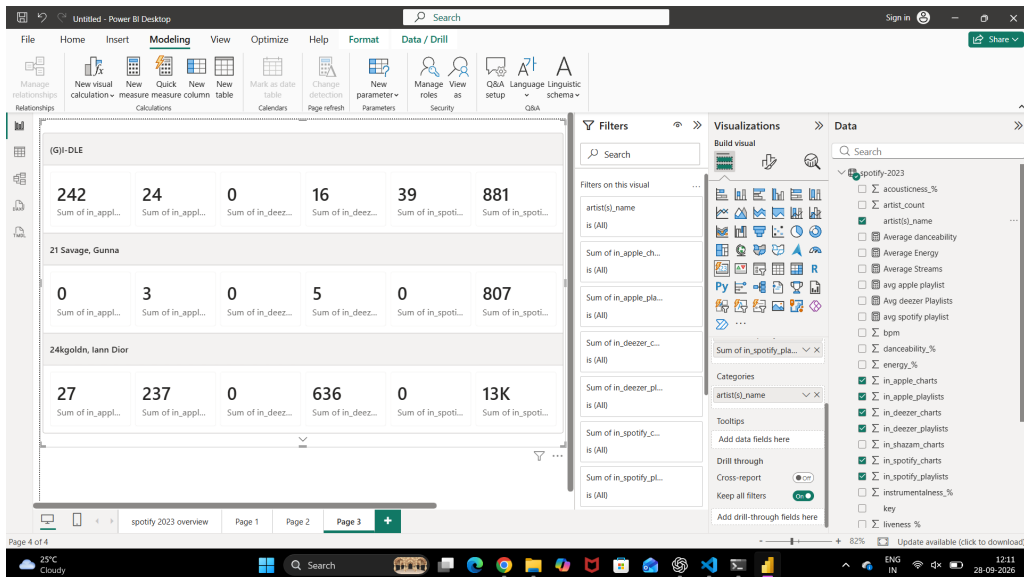


Figure 19: Multi-row card of platform chart and playlist counts by artist

## 6.4 Page 4: Spotify vs Apple Charts (Slicer and Bar Chart)

A tile-style slicer on **artist(s)\_name** controls a bar chart comparing **Sum of in\_spotify\_charts** with **Sum of in\_apple\_charts**. Selecting *Jimin* shows a longer Apple bar, with Apple chart appearances of about 250 against roughly 100 on Spotify. Selecting *Coldplay* reverses the pattern, with Spotify at about 70 against roughly 25 on Apple. This lets a viewer see quickly which platform favors each artist.

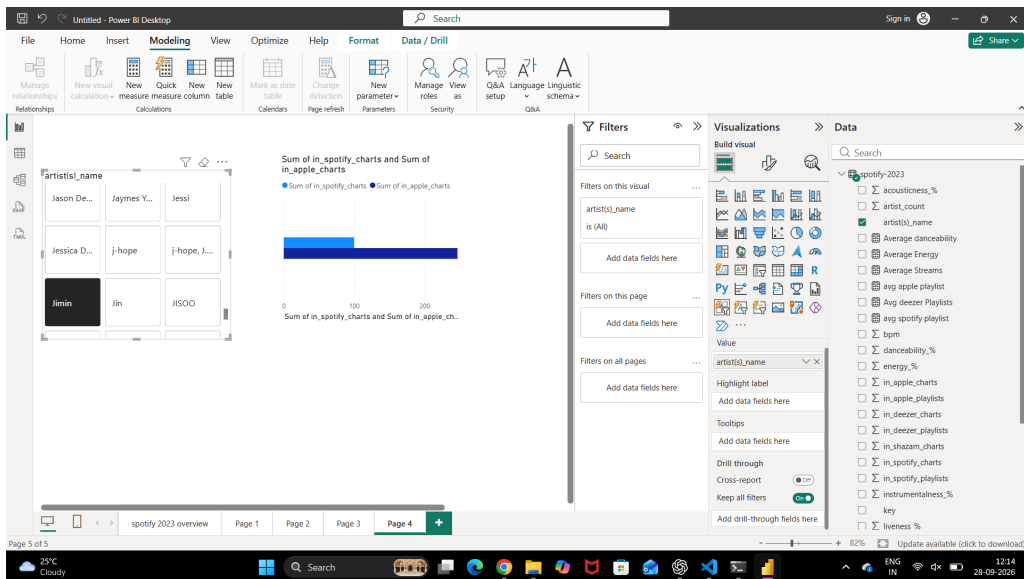


Figure 20: Jimin selected: stronger performance on Apple charts

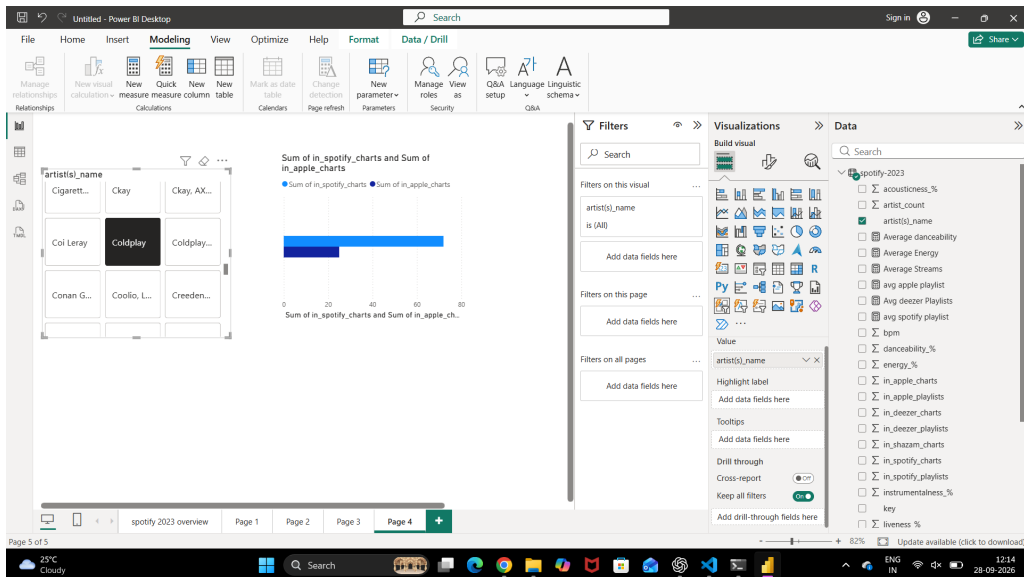


Figure 21: Coldplay selected: stronger performance on Spotify charts

## 6.5 Page 5: Release Month Filter

A numeric range slicer on **released\_month** (set here to months 1 to 7) is paired with a list of **track\_name** values. Moving the slider narrows the list to songs released within the chosen months, making it easy to explore the dataset by release period.

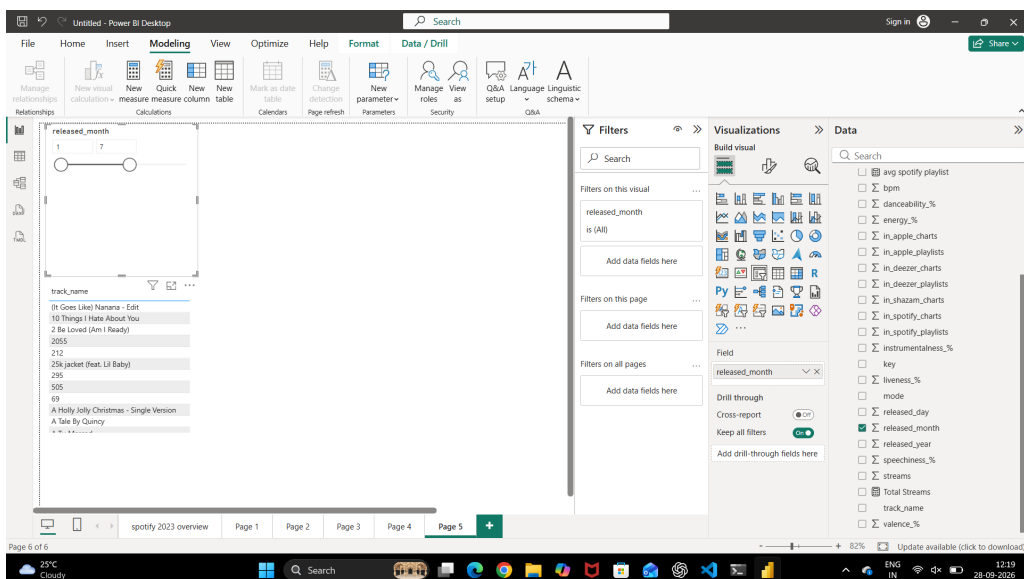


Figure 22: released\_month slider with the track\_name list

## 7. Key Findings

- A few tracks, led by Blinding Lights and Shape of You, account for streaming totals far above the rest of the dataset.
- Total streams are highly concentrated: a handful of artists reach 13 to 14 billion streams while most sit at a small fraction of that.
- Platform preference varies by artist: Jimin performs stronger on Apple charts, while Coldplay performs stronger on Spotify charts.

- Spotify playlist reach is far larger than Apple or Deezer for the artists shown on the platform card, for example 24kgoldn, Iann Dior.

## 8. Conclusion

Power Query was used to remove duplicates, clean text and blank rows, reducing the dataset to 943 records. DAX measures provided reusable averages, and five report pages with bar charts, bubble charts, cards and slicers made the data interactive. The report shows that streaming success is concentrated in a small set of tracks and artists and that chart performance differs between Spotify and Apple.